Visual Intelligence

REVIEW Open Access

State space model for new generation network alternative to transformers: a survey

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Abstract

In the post-deep learning era, the Transformer architecture has demonstrated its powerful performance in pre-trained large models and various downstream tasks. However, the enormous computational demands of Transformers have deterred many researchers. To reduce the complexity of attention models, numerous efforts have been made to design more efficient methods. Among them, the state space model (SSM), as a possible replacement for the self-attention based Transformer model, has drawn more and more attention in recent years. This paper provides the first comprehensive review of SSM works and offers experimental comparisons and analyses to better demonstrate SSM's features and advantages. First, we provide a detailed description of the principles to help the readers quickly grasp the key ideas of the SSM. Then, we will review existing SSMs and their various applications, including those in the fields of natural language processing, computer vision, graphs, multi-modal and multi-media, point cloud and event stream, and other domains. Next, we provide statistical comparisons and analyses of these models to help readers understand how different structures affect performance on various tasks. Finally, we suggest potential research directions that could further advance the development of the theoretical model and propose possible research points in this direction to better promote the application of SSMs.

Keywords: State space model, Mamba, Transformer, Linear attention, Computer vision, Natural language processing

1 Introduction

Artificial intelligence (AI) has experienced rapid development since the early 2010s, a period often referred to as the third wave of AI. During this era, connectionism-based deep learning technology has played an extremely important role. The origins of deep learning can be traced back to AlexNet's proposal [1], which achieved the best performance in the ImageNet [2] competition, far surpassing the second-place result. Various convolutional neural networks (CNNs) were then proposed, such as VGG [3], ResNet [4] and GoogleNet [5]. The concepts of blocks, residual connections, and inception have inspired the design of many subsequent deep neural networks [6, 7]. On the other hand, the family of recurrent neural networks (RNNs), such as long short-term memory (LSTM) [8] and gated recurrent unit (GRU) [9], dominates the sequence-based learning community, including natural language processing (NLP), and audio processing. Graph neural networks (GNNs) [10, 11] are proposed to broaden the applications of deep neural networks on graph data. Even with maximum support in terms of datasets and computing power, however, these mainstream models still encounter bottlenecks.

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To address the limitation of CNN-, RNN-, and GNN-based models being primarily constrained to local relation modeling, the Transformer [12], which was proposed in 2017, effectively learns long-range feature representations. The core operation is the self-attention mechanism, which transforms the input tokens into query, key, and value features. It then outputs the long-range features by multiplying a similarity matrix obtained by multiplying the query and key features by the value features. The Transformer architecture first gained popularity in the NLP community with the help of *pre-training and fine-tuning* paradigm [13], as demonstrated by models such as BERT [14], BART [15], ERNIE [16], and GPT [17]. Then, other communities are also boosted with such networks. For example, the ViT [18] and Swin-Transformer [19] were released in computer vision. Many researchers exploit the hybrid network architectures by combining Transformers with other networks, or adapting Transformers for multi-modal research problems [20, 21]. Large foundation models are currently emerging, and parameter-efficient fine-tuning (PEFT) strategies [22] have advanced significantly as well. However, current Transformer-based models still require high-end graphics cards with ample memory for training, testing, and deployment, which greatly limits their wider application.

To decrease computing costs, while capturing long-range dependencies and maintaining high performance, many new sparse, attention-based models or new neural network paradigms have been proposed [23–27]. Among them, the state space model (SSM) (e.g., Mamba [28], S4 [29], and S4ND [30]) has become the center of attention. The SSM is a framework that was initially proposed for modeling dynamic systems using state variables in fields such as control theory and computational neuroscience.[1] When adapting this concept for deep learning, we usually refer to linear invariant (or stationary) systems. The original SSM is a continuous dynamic system that can be converted into a discrete form for the computer to handle, such as recurrent or convolutional views. SSMs can be adopted for various data processing and feature learning, including image and video data, text data, structured graph data, event streams and point cloud data, multi-modal and multi-media data, audio and speech, time series data, and tabular data. The SSM can also be utilized to build efficient generative models, such as SSM-based diffusion generative models [31–33].

There has been a growing focus on the study of SSMs, the timeline of which is shown in Fig. 1. In this paper, we first introduce the working principle of the SSM in Sect. 2. Then, in Sect. 3, we review related works on SSMs from multiple perspectives, including the origin and variation of SSMs, and SSM models and applications in different domains. More importantly, we conduct extensive experiments on multiple downstream tasks to validate the effectiveness of SSMs in Sect. 4. Downstream tasks involve single- and multi-label classification, visual object tracking, pixel-level segmentation, image-to-text generation, and person or vehicle re-identification. We also propose several possible research directions for the theory and applications of SSMs in Sect. 6. Finally, we present our conclusions in Sect. 7.

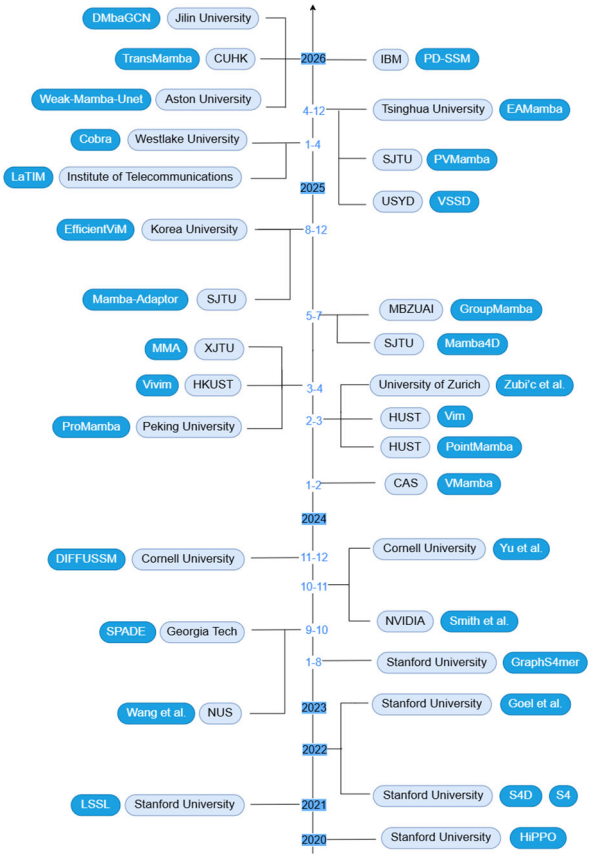


Figure 1 The timeline of representative state space model (SSM)-based algorithms published from 2020 to July of 2026. These methods are HiPPO [34], LSSL [35], Goel et al. [36], S4D [37], S4 [29], GraphS4mer [38], SPADE [39], Wang et al. [40], Yu et al. [41], Smith et al. [42], DIFFUSSM [31], VMamba [43], Zubić et al. [44], Vim [45], PointMamba [46], MMA [47], Vivim [48], ProMamba [49], GroupMamba [50], Mamba4D [51], EfficientViM [52], Mamba-Adaptor [53], Cobra [54], LaTIM [55], EAMamba [56], PVMamba [57], VSSD [58], DMbaGCN [59], TransMamba [60], Weak-Mamba-Unet [61], and PD-SSM [62]. Georgia Tech: Georgia Institute of Technology; NUS: National University of Singapore; CAS: Chinese Academy of Sciences; HUST: Huazhong University of Science and Technology; XJTU: Xi'an Jiaotong University; HKUST: The Hong Kong University of Science and Technology; MBZUAI: Mohamed bin Zayed University of Artificial Intelligence; SJTU: Shanghai Jiao Tong University; USYD: The University of Sydney; CUHK: The Chinese University of Hong Kong

[1] https://huggingface.co/blog/lbourdois/get-on-the-ssm-train.

2 Formulation of SSM

State space model (SSM) originates from the classic Kalman filter [63]. It takes the 1-D input signal $\boldsymbol{U}(t)$ and maps it into N-D latent state $\boldsymbol{X}(t)$. Then, it projects into a 1-D output signal $\boldsymbol{y}(t)$. The general computing procedure can be defined in Eq. (1):

$$\begin{aligned} \dot{\boldsymbol{X}}(t) &= \boldsymbol{A}(t)\boldsymbol{X}(t) + \boldsymbol{B}(t)\boldsymbol{U}(t), \\ \boldsymbol{y}(t) &= \boldsymbol{C}(t)\boldsymbol{X}(t) + \boldsymbol{D}(t)\boldsymbol{U}(t), \end{aligned} \tag{1}$$

where $\boldsymbol{X}(t) \in \mathbb{R}^n$, $\boldsymbol{y}(t) \in \mathbb{R}^q$, and $\boldsymbol{U}(t) \in \mathbb{R}^p$ denote the state vector, output vector, and input (or control) vector, respectively. Here, t represents the continuous time index, while n, q, and p denote the dimensions of the state, output, and input spaces, respectively. The notation $\dot{\boldsymbol{X}}(t) = \frac{d}{dt}\boldsymbol{X}(t)$ represents the temporal derivative of the state vector, where d is the differential operator. $\boldsymbol{A}(t) \in \mathbb{R}^{n\times n}$, $\boldsymbol{B}(t) \in \mathbb{R}^{n\times p}$, $\boldsymbol{C}(t) \in \mathbb{R}^{q\times n}$, and $\boldsymbol{D}(t) \in \mathbb{R}^{q\times p}$ represent the state transition matrix, input matrix, output matrix, and feed-forward matrix, respectively. When there is no direct feedthrough in the system model, $\boldsymbol{D}(t)$ is a zero matrix. Thus, we obtain the following simplified equations:

$$\begin{aligned} \dot{\boldsymbol{X}}(t) &= \boldsymbol{A}(t)\boldsymbol{X}(t) + \boldsymbol{B}(t)\boldsymbol{U}(t), \\ \boldsymbol{y}(t) &= \boldsymbol{C}(t)\boldsymbol{X}(t). \end{aligned} \tag{2}$$

As the raw system is continuous, we need to first discretize it before feeding the computer. For the Mamba architecture, the zero-order hold (ZOH)[2] is adopted for the discretization, and we have

$$\begin{aligned} \boldsymbol{X}_t &= \overline{\boldsymbol{A}}\boldsymbol{X}_{t-1} + \overline{\boldsymbol{B}}\boldsymbol{U}_t, \\ \boldsymbol{y}_t &= \boldsymbol{C}\boldsymbol{X}_t, \end{aligned} \tag{3}$$

where $\overline{\boldsymbol{A}} = exp(\Delta\boldsymbol{A})$, $\overline{\boldsymbol{B}} = (\Delta\boldsymbol{A})^{-1}(exp(\Delta\boldsymbol{A}) - \boldsymbol{I}) \cdot \Delta\boldsymbol{B}$, $\boldsymbol{I}$ is the identity matrix and Δ denotes the step size. If we denote the state vector and input vector using $\boldsymbol{h}$ and $\boldsymbol{x}$, we obtain the following functions similar to the computing procedure of the recurrent neural network (RNN) model:

$$\begin{aligned} \boldsymbol{h}_t &= \overline{\boldsymbol{A}}\boldsymbol{h}_{t-1} + \overline{\boldsymbol{B}}\boldsymbol{x}_t, \\ \boldsymbol{y}_t &= \boldsymbol{C}\boldsymbol{h}_t. \end{aligned} \tag{4}$$

However, similar to the RNN model, we face the dilemma that the computation cannot be parallelized. By simply expanding Eq. (4), we have

$$\begin{aligned} \boldsymbol{y}_0 &= \boldsymbol{C}\bar{\boldsymbol{A}}^0\bar{\boldsymbol{B}}x_0, \\ \boldsymbol{y}_1 &= \boldsymbol{C}\bar{\boldsymbol{A}}^1\bar{\boldsymbol{B}}x_0 + \boldsymbol{C}\bar{\boldsymbol{A}}^0\bar{\boldsymbol{B}}x_1, \\ \boldsymbol{y}_2 &= \boldsymbol{C}\bar{\boldsymbol{A}}^2\bar{\boldsymbol{B}}x_0 + \boldsymbol{C}\bar{\boldsymbol{A}}^1\bar{\boldsymbol{B}}x_1 + \boldsymbol{C}\bar{\boldsymbol{A}}^0\bar{\boldsymbol{B}}x_2. \end{aligned} \tag{5}$$

[2] https://en.wikipedia.org/wiki/Zero-order_hold.

We note that the multipliers of the last and penultimate terms are always $\boldsymbol{C}\bar{\boldsymbol{A}}^0\bar{\boldsymbol{B}}$ and $\boldsymbol{C}\bar{\boldsymbol{A}}^1\bar{\boldsymbol{B}}$. Therefore, we treat these multipliers as the convolutional kernel $\overline{\boldsymbol{K}} = \boldsymbol{C}\bar{\boldsymbol{B}} \cdot (\bar{\boldsymbol{A}}^0, \bar{\boldsymbol{A}}^1, \bar{\boldsymbol{A}}^2, \ldots, \bar{\boldsymbol{A}}^L)$, where L is the length of the given input sequence. We can rewrite Eq. (4) as the following convolutional equations:

$$\begin{aligned} \overline{\boldsymbol{K}} &= (\boldsymbol{C}\overline{\boldsymbol{B}}, \boldsymbol{C}\overline{\boldsymbol{A}\boldsymbol{B}}, \ldots, \boldsymbol{C}\overline{\boldsymbol{A}}^k\overline{\boldsymbol{B}}, \ldots), \\ \boldsymbol{y} &= x * \overline{\boldsymbol{K}}. \end{aligned} \tag{6}$$

At this moment, the complete SSM model can realize the parallelism of training and is suitable for the recurrent form of linear complexity of inference. In the Transformer architecture, the context information is stored in the similarity matrix. However, the SSM does not have a similar module, which makes it perform poorly in contextual learning.

To address this issue, the Mamba architecture, proposed by Gu and Dao [28], improves upon the following two aspects: 1) The selective scan operator enables the model to filter out irrelevant information. In practical implementation, the Δ, $\boldsymbol{B}$, and $\boldsymbol{C}$ become functions of the input; meanwhile, the matrix $\boldsymbol{A}$ remains unchanged. 2) The hardware-aware algorithm enables the efficient storage of intermediate results through parallel scanning, kernel fusion, and recalculation. The prevalence of the SSM or Mamba architectures can be attributed to their distinctive characteristics.

3 State space model

In this section, we review the related works on the SSM architectures and applications. We divide the related works into the following domains, i.e., the origin and variation of the SSM, natural language processing, computer vision, graphs, multi-modal and multi-media, point cloud and event stream, and others.

3.1 Origin and variation of SSM

The SSM originates from classical state estimation and dynamical system modeling. Kalman filtering can be divided into prediction and correction steps. The SSM is a mathematical model that describes the behavior of a dynamic system using a set of first-order differential equations (continuous time systems) or difference equations (discrete time systems) to represent the evolution of the internal state of the system, and another set of equations to describe the relationship between the state and the output of the system. This formulation provides a compact representation of temporal dynamics and naturally supports recurrent computation. However, it is challenging to use general state space parameterizations directly in deep neural networks because the dynamics may be difficult to optimize and may fail to preserve information over long time

Table 1 Summary of foundational state space model (SSM)-based models

#ID	Algorithm	Source	Domain	Parameters	Architecture	Efficiency	Code
1	HiPPO [34]	NeurIPS20	Origin	N/A	N/A	Speed: 470,000 elements/s	https://github.com/state-spaces/s4
2	Gu et al. [35]	NeurIPS21	Origin	N/A	LSSL	N/A	N/A
3	S4 [29]	ICLR22	Origin	249M	DPLR	Memory allocation: 33.5 MB Training step time: 4.75 ms	https://github.com/state-spaces/s4
4	S4D [37]	NeurIPS22	Origin	N/A	Diagonal	N/A	N/A
5	DSS [64]	NeurIPS22	Origin	N/A	SSM	N/A	https://github.com/ag1988/dss
6	LRU [65]	ICML23	Origin	N/A	RNNs, SSM	Speed: 15.9 steps/s	N/A
7	S4-LegS [66]	ICLR23	Origin	150K	S4, Math	N/A	N/A

periods. A summary of foundational SSM-based models is presented in Table 1.

Improving long-range memory through structured dynamics Similar to RNNs, SSMs can suffer from vanishing and exploding gradients when modeling long sequences. HiPPO [34] combines recurrent memory with optimal polynomial projections to improve long-range memory. The process is described by Eq. (7).

$$\boldsymbol{A}_{nk} = \begin{cases} (2n+1)^{\frac{1}{2}}(2k+1)^{\frac{1}{2}} & \text{if } n > k \\ n+1 & \text{if } n = k \\ 0 & \text{if } n < k \end{cases}, \tag{7}$$

where n and k indicate the row and column indices of $\boldsymbol{A}$.

From dynamical systems to learnable sequence layers Gu et al. [35] introduced the linear state space layer (LSSL), combining recurrent, convolutional, and differential equation perspectives while enabling recurrent inference and parallel convolutional training. Based on this foundation, Gu et al. [29] proposed S4, a structured parameterization that captures long-range dependencies with improved efficiency. S4 preserves information over long intervals but does not explicitly select input-dependent information.

Simplifying structured state space parameterizations The complexity of S4 motivated a second line of research that simplifies the state transition structure. Gu et al. [37] showed that diagonal state spaces such as S4D can match S4 without low-rank corrections while simplifying implementation. DSS [64] similarly uses diagonal state matrices to obtain efficient long-range sequence modeling. Their limited interactions can reduce dynamical expressiveness. The linear recurrent unit (LRU) [65] transfers these ideas to recurrent architectures and supports efficient long context processing. S4-LegS [66] studies efficient Legendre-based structured dynamics for long-range sequence modeling. Based on the origin and variations of the SSM mentioned above, many SSM-based works are constantly emerging, including but not limited to natural language processing, computer vision, and more. The following summaries will introduce the expansion and application of various fields.

3.2 Natural language processing

In recent years, the development of large language models has revolutionized the field of natural language processing (NLP). However, the widely used Transformer architecture is limited by high computational and memory requirements. A summary of recent SSM-based models for NLP is presented in Table 2.

For speech separation, DPMamba [67] replaces the conventional Transformer with selective SSM blocks to model short- and long-term bidirectional dependencies. DenseMamba [68] integrates hidden states from preceding layers to retain shallow information and reduce losses caused by repeated state transitions. LaTIM [55] decomposes Mamba's latent computation into token-level interactions to reveal how information propagates through the model.

In the translation task, Ref. [69] formulates the problem of generating dance choreography as a translation task and proposes the MDLT, which utilizes existing datasets to learn how to translate audio sequences into corresponding dance poses. TransMamba [60] develops this hybrid direction through a sequence-level Mamba–Transformer architecture, combining the contextual modeling ability of attention with the linear-complexity sequence modeling of Mamba.

3.3 Computer vision

Recently, the linear time series modeling of the SSM has attracted widespread attention, demonstrating strong performance in the NLP field. Inspired by these progress, many SSM-based vision models have been proposed for image classification, object detection, segmentation, medical image analysis, image restoration, generation, video understanding, visual tracking, and other visual tasks.

Table 2 Summary of recent SSM-based models for natural language processing (NLP)

#ID	Algorithm	Source	Domain	Parameters	Architecture	Efficiency	Code
1	DPMamba [67]	ICASSP25	NLP	DPMamba-XS 2.3M DPMamba-S 8.1M DPMamba-M 15.9M DPMamba-L 59.8M	Mamba	N/A	N/A
2	DenseMamba [68]	NAACL25	NLP	DenseMamba-360M DenseMamba-1.3B	Mamba	N/A	https://github.com/WailordHe/DenseSSM
3	LATIM [55]	ACL25	NLP	13M 130M–1.4B	Mamba	N/A	https://github.com/deep-spin/latim
4	MDLT [69]	NCA26	NLP	N/A	Mamba	N/A	https://github.com/meowatthemoon/MDLT
5	TransMamba [60]	AAAI26	NLP	TransMamba-400M TransMamba-1.5B	Mamba+ Transformer	FLOPs/Layer: 1.91×10^{10}	https://github.com/Yixing-Li/TransMamba

A summary of recent SSM-based models for computer vision is presented in Table 3. VM-UNet [70] establishes a pure SSM baseline for medical image segmentation, while nnMamba [71] combines convolutional detail extraction with long-range state space modeling for 3D medical images. Pan-Mamba [72] applies channel-swapping and cross-modal Mamba modules to pan-sharpening. EfficientVMamba [73] adds convolutional branches for efficient visual representation, and VmambaIR [74] extends linear-complexity state space modeling to image restoration.

H-vmunet [75] enhances local feature learning through high-order interactions. Gamba [76] uses a Mamba sequence network for efficient single-view 3D reconstruction. HARMamba [77] targets lightweight wearable sensor activity recognition, and UltraLight VM-UNet [78] develops an extremely compact U-shaped medical segmentation model.

For spatio-temporal and tracking tasks, Mamba4D [51] decouples spatial and temporal state modeling for long point-cloud videos. Vamba [79] encodes video tokens with Mamba-2 blocks. TemTrack [80] separates temporal dynamics from appearance modeling, while MamTrack [81] introduces lightweight gate-driven multimodal fusion.

For detection and classification, SAMamba [82] combines SAM2 with selective state space modeling for infrared small target detection. M3Net [83] uses modality-adaptive fusion and channel scaling for autonomous driving tasks. ECP-Mamba [84] combines multi-scale self-supervised contrastive learning with SSMs for PolSAR classification. UD-Mamba [85] introduces uncertainty-driven sequential and skip scanning, and Mamba-Sea [86] combines global and local sequence enhancement for cross-domain medical segmentation. MTTM [87] uses a memory-augmented processing unit, while MobileViM [88] combines bidirectional traversal with cross-scale bridging for medical images.

More recent backbones and task-specific models broaden this scope. DefMamba [89] learns deformable scan paths, Mamba-Adaptor [53] adds task-oriented spatial adaptors, 2DMamba [90] extends selective modeling to two dimensions, and GroupMamba [50] uses grouped state space processing. MambaVision [91] combines hierarchical Mamba and Transformer blocks, EfficientViM [52] emphasizes computational efficiency, and PVMamba [57] parallelizes visual Mamba through dynamic state aggregation.

MambaVLT [92] exploits temporal state evolution for vision-language tracking. VSSD [58] introduces non-causal state space duality for visual representation. EAMamba [56] uses multi-head selective scanning for image restoration, and Mamba-3VL [93] models spatial and cross-modal relationships in 3D vision-language learning. MambaML [94] captures visual and label dependencies for multi-label classification. Wave-MambaAD [95] incorporates wavelet-driven state space modeling for anomaly detection, and Semi-ViM [96] addresses label imbalance in semi-supervised learning.

Weak-Mamba-Unet [61] learns dense pseudo-labels for weakly supervised medical segmentation. VideoMamba [97] evaluates Mamba as an efficient alternative for diverse video understanding tasks. JambaTalk [98] combines Mamba, Transformer, and mixture-of-experts components for controllable generation.

3.4 Graphs

In addition to standard grid data, graph-structured data such as social networks and protein structures are widely studied. Such data can be transformed into ordered representations, enabling SSMs to capture long-range dependencies efficiently. A summary of recent SSM-based graph models is presented in Table 4. GG-SSM [99] dynamically constructs graphs using feature relationships and Chazelle's minimum spanning tree algorithm, enabling

Table 3 Summary of recent SSM-based models for computer vision (CV)

#ID	Algorithm	Source	Domain	Parameters	Architecture	Efficiency	Code
1	VM-UNet [70]	ACM TOMM25	CV	30M	SSM+CNN	N/A	https://github.com/JCruan519/VM-UNet
2	nnMamba [71]	ISBI25	CV	15.55M	SSM+CNN	GFLOPs: 141.14	https://github.com/lhaof/nnMamba
3	Pan-Mamba [72]	Information Fusion25	CV	0.1827M	Mamba	GFLOPs: 3.0088	https://github.com/alexhe101/Pan-Mamba
4	EfficientV-Mamba [73]	AAAI25	CV	Tiny: 6M Small: 11M Base: 33M	SSM	N/A	https://github.com/TerryPei/EfficientVMamba
5	VmambaIR [74]	TCSVT25	CV	10.5M	SSM	N/A	N/A
6	H-vmunet [75]	Neurocomputing25	CV	N/A	VM-UNet	N/A	https://github.com/wurenkai/H-vmunet
7	Gamba [76]	TPAMI25	CV	N/A	Mamba	N/A	N/A
8	HARMamba [77]	IoTJ25	CV	0.7966M (PAMAP2) 0.7880M (UCI) 0.7883M (UNIMIB HAR) 0.7877M (WISDM)	Mamba	PAMAP2: 0.27921GFLOPs UCI: 0.23783GFLOPs UNIMIBHAR: 0.23836GFLOPs WISDM: 0.25652GFLOPs	N/A
9	UltraLight VM-UNet [78]	Patterns25	CV	0.049M	VMamba+CNN	GPU memory: 32 GB GFLOPs: 31.65	https://github.com/wurenkai/UltraLight-VM-UNet
10	Mamba4D [51]	CVPR25	CV	N/A	SSM	N/A	https://github.com/IRMVLab/Mamba4D
11	Vamba [79]	ICCV25	CV	N/A	Mamba-2	N/A	https://github.com/TIGER-AI-Lab/Vamba
12	TemTrack [80]	AAAI25	CV	70M	Mamba	GFLOPs: 24.8 FPS: 46	https://github.com/GXNU-ZhongLab/TemTrack
13	MamTrack [81]	CVPR25	CV	N/A	Mamba	FPS: 18.1	https://github.com/scy0712/MamTrack
14	SAMamba [82]	Information Fusion25	CV	Hiera-T 28.65M Hiera-S 36.34M	Mamba	Hiera-T FPS: 7.32 Hiera-S FPS: 6.82	https://github.com/zhengshuchen/SAMamba
15	M3Net [83]	AAAI25	CV	N/A	Mamba	N/A	https://github.com/Cedarch/M3Net
16	ECP-Mamba [84]	TGRS25	CV	4.36M	Mamba	N/A	https://github.com/HaixiaBi1982/ECP-Mamba
17	UD-Mamba [85]	arXiv25	CV	19.12M	Mamba	GFLOPs: 5.91	https://github.com/piooip/UD-Mamba
18	Mamba-Sea [86]	IEEE TMI25	CV	27.43M	Mamba	GFLOPs: 65.37	https://github.com/orange-czh/Mamba-Sea
19	MTTM [87]	ICASSP25	CV	31M	Mamba	N/A	https://github.com/hongkai-wei/MTTM
20	MobileViM [88]	arXiv25	CV	MobileViM-xs 2.89M MobileViM-s 6.29M	Mamba	N/A	https://github.com/anthonyweidai/MobileViM_3D
21	DefMamba [89]	CVPR25	CV	T: 8M S: 32M B: 51M	Deformable Mamba	T: 1.2G FLOPs S: 4.8G FLOPs B: 8.5G FLOPs	N/A

Table 3 (*Continued*)

#ID	Algorithm	Source	Domain	Parameters	Architecture	Efficiency	Code
22	Mamba-Adaptor [53]	CVPR25	CV	b1: 7.8M b2: 32M	Mamba+ Adaptor	b1: 1.4G FLOPs b2: 5.4G FLOPs	N/A
23	2DMamba [90]	CVPR25	CV	Cls: 30M Seg: 62M	2D SSM	Classification: 4.94G FLOPs Segmentation: 950G FLOPs	https://github.com/AtlasAnalyticsLab/2DMamba
24	Group-Mamba [50]	CVPR25	CV	T: 23M S: 34M B: 57M	Group SSM	T: 4.5G FLOPs S: 7.0G FLOPs B: 14.0G FLOPs	https://github.com/Amshaker/GroupMamba
25	MambaVision [91]	CVPR25	CV	97.7M	Mamba+ Transformer	FLOPs: 15.0G Throughput: 3670 images/s	https://github.com/NVlabs/MambaVision
26	EfficientViM [52]	CVPR25	CV	19.6M	HSM-SSD	FLOPs: 1.11G Throughput: 8170 images/s	https://github.com/mlvlab/EfficientViM
27	PVMamba [57]	ICCV25	CV	24M–89M	SSM+DSA	FLOPs: 4.5G–16.1G	https://github.com/VISION-SJTU/PVMamba
28	MambaVLT [92]	CVPR25	CV	149M	Multimodal SSM	FLOPs: 70G	-
29	VSSD [58]	ICCV25	CV	M: 14M T: 24M S: 40M B: 89M	Non-Causal SSD	M: 2.3G FLOPs T: 4.5G FLOPs S: 7.4G FLOPs B: 16.1G FLOPs	https://github.com/YuHengsss/VSSD
30	EAMamba [56]	ICCV25	CV	25.3M	Mamba-Former+ MHSSM	FLOPs: 137G	https://github.com/daidaijr/EAMamba
31	Mamba-3VL [93]	ICCV25	CV	N/A	Mamba Mixer+IDPA	N/A	https://github.com/wangyuan123ac/Mamba-3VL
32	MambaML [94]	ICCV25	CV	85.0M	Bi-Mamba+ Mamba Decoder	FLOPs: 36.9G Input: 448 × 448	N/A
33	Wave-MambaAD [95]	ICCV25	CV	22.3M	Wavelet-Mamba	FLOPs: 7.5G Memory: 1142 MB Speed: 58.8	N/A
34	Semi-ViM [96]	ICCV25	CV	T: 12M S: 27M B: 146M	Bidirectional SSM	N/A	N/A
35	Weak-Mamba-Unet [61]	TBME26	CV	N/A	CNN ViT Mamba	N/A	https://github.com/ziyangwang007/Mamba-UNet
36	Video Mamba Suite [97]	IJCV26	CV	ViViM-T: 7M ViViM-S: 26M	SSM+ViT	N/A	https://github.com/OpenGVLab/video-mamba-suite
37	JambaTalk [98]	ACM TOMM26	CV	N/A	Mamba	N/A	N/A

Table 4 Summary of recent SSM-based models for graphs

#ID	Algorithm	Source	Domain	Parameters	Architecture	Efficiency	Code
1	GG-SSM [99]	CVPR25	Graph	GG-SSM 0.22M	SSM+Graph	GFLOPs: 8.01	https://github.com/uzh-rpg/gg_ssms
2	DyG-Mamba [100]	NeurIPS25	Graph	N/A	SSM	GPU Usage: 18094M	https://github.com/Clearloveyuan/DyG-Mamba
3	MGLC [101]	AAAI26	Graph	N/A	CNN+SSM	YouTubeFace Training time: 1.28 h	https://github.com/zhangyuanyang21/2026-AAAI-MGLC
4	DMbaGCN [59]	AAAI26	Graph	N/A	GNN+Mamba	N/A	https://github.com/hexin5515/DMbaGCN

adaptive modeling of complex dependencies in sequential and high-dimensional data. DyG-Mamba [100] extends state space modeling to dynamic graphs by incorporating irregular temporal intervals into continuous state evolution and selectively retaining useful historical information.

MGLC [101] employs Mamba-based sequence modeling to jointly capture cross-view dependencies and global-local structural relationships for multi-view clustering. DMbaGCN [59] integrates local state evolution and global context-aware Mamba modules into graph convolutional networks to preserve node-specific representations and alleviate over-smoothing.

3.5 Multi-modal and multi-media

The SSM can also be adapted to multimodal and multimedia tasks. A summary is presented in Table 5. S5 [42] simplifies structured state space layers, while gated RNNs [102] connect recurrent computation with linear self-attention. Mamba [28] introduces input-dependent selection, and Grazzi et al. [103] evaluated its in-context learning behavior. MambaFormer [104] combines Mamba with attention to improve retrieval-oriented tasks.

MambaMIL [105] captures long-range dependencies between dispersed pathology instances. Motion Mamba [106] combines hierarchical temporal and bidirectional spatial Mamba for motion generation. VL-Mamba [107] reduces the cost of multimodal language modeling, and CMViM [108] learns aligned representations for high-resolution multimodal medical images. Decision Mamba [109] applies Mamba to offline reinforcement learning, while EMMA [110] uses pixel-wise alignment and multi-scale fusion for multimodal language modeling.

Ali et al. [111] analyzed the implicit attention behavior of selective state space layers. Cobra [54] replaces the conventional Transformer language backbone with Mamba for multimodal inputs, and Sigma [112] applies Mamba to multimodal semantic segmentation. OmniMamba [113] unifies text and image generation through next-token prediction. AVS-Mamba [114], LEAF-Mamba [115], and VSumMamba [116] address audio-visual segmentation, RGB-D salient-object detection, and video summarization, respectively. FluencyVE [117] improves long-range temporal modeling for video editing, and MSF-Mamba [118] selectively aggregates temporal states for micro-gesture recognition.

3.6 Event stream and point cloud data

We summarize recent SSM-based point-cloud and event-stream algorithms in Table 6.

PointMamba [46] reorders embedded point patches and processes them with Mamba blocks for linear-complexity global modeling. Point Mamba [119] uses octree-based ordering to preserve spatial proximity and causal dependence. 3DMambaComplete [120] introduces HyperPoint generation and spreading for point-cloud completion. Zubi'c et al.'s method [44] uses learnable time scales to adapt event-based models to different input frequencies. PCM [121] converts point clouds into consistent traversal sequences and supplies order prompts. 3DMambaIPF [122] combines Mamba with a differentiable rendering loss for point-cloud filtering.

EventMamba [123] treats events as pseudo-point clouds for classification and regression, while the reconstruction-oriented EventMamba [124] preserves spatio-temporal locality through randomized windows and consistent traversal. SMamba [125] introduces adaptive sparsification for event-based detection. StruMamba3D [126] learns structural point-cloud representations, PMA [127] fuses intermediate features from pretrained models, and SI-Mamba [128] incorporates spectral information for robust processing. Mamba4D [51] models long spatio-temporal point-cloud sequences. MambaEVT [129] uses dynamic template updating for event tracking. DAPointMamba [130] addresses domain-adaptive completion, and CloudMamba [131] combines grouped selective state spaces with improved serialization.

3.7 Others

We summarized other SSM-based algorithms in different domains in Table 7, including series forecasting and physics-informed modeling, general sequence modeling, architecture design, and SSM expressiveness, physics-informed modeling and dynamical systems, and applications in biomedical and bioinformatics, and speech and audio processing. SaShiMi [36] is a multi-scale S4 architecture for raw audio generation. Lu et al. [132] modified S5 to reset hidden states within variable-length trajectories. David et al. [133] combined state space dynamics with probabilistic hidden states. FlashFFTConv [134] accelerates long convolutions with hardware-aware FFT kernels. CLDSSM [135] applies regularization-based continual learning to dynamic systems, and LaRocque et al. [136] evaluated Mamba for proprioceptive terrain classification.

SiMBA [137] combines Mamba sequence modeling with EinFFT channel modeling. Sharma et al. [138] investigated factual recall in Mamba models. Forgione et al. [139] reduced the order of learned linear dynamic blocks while preserving essential dynamics. RecMamba [140] models evolving user preferences for efficient recommendation, and SpatialNet-Mamba [141] applies SSMs to multi-channel speech enhancement. Caduceus [142] provides bidirectional, reverse-complement-equivariant DNA sequence modeling.

MambaStock [143] models nonlinear stock dynamics with S4. HiSS [144] constructs a multi-resolution temporal hierarchy for continuous sequence prediction, while

Table 5 Summary of recent SSM-based multi-modal and multi-media algorithms

#ID	Algorithm	Source	Domain	Parameters	Architecture	Task	Code
1	S5 [42]	ICLR23	Multi-modal Multi-media	280K	SSM	Classification	https://github.com/lindermanlab/S5
2	Zucchet et al. [102]	arXiv23	Multi-modal Multi-media	N/A	N/A	N/A	N/A
3	Mamba [28]	arXiv23	Multi-modal Multi-media	Mamba-130M Mamba-370M Mamba-790M Mamba-1.4B	Mamba	Synthetic Zero-shot	https://github.com/state-spaces/mamba
4	Grazzi et al. [103]	AutoML24	Multi-modal Multi-media	N/A	N/A	N/A	N/A
5	MambaFormer [104]	ICML24	Multi-modal Multi-media	N/A	Mamba+ Former	N/A	N/A
6	MambaMIL [105]	MICCAI24	Multi-modal Multi-media	N/A	Mamba	Survival Prediction Cancer Subtyping	https://github.com/isyangshu/MambaMIL
7	Motion Mamba [106]	arXiv24	Multi-modal Multi-media	N/A	Mamba	Motion Synthesis Text-to-Motion	https://github.com/steve-zeyu-zhang/MotionMamba
8	VL-Mamba [107]	NeurIPS-W24	Multi-modal Multi-media	Mamba LLM 2.8B	Mamba	Multimodal Learning	https://yanyuanqiao.github.io/vl-mamba
9	CMViM [108]	arXiv24	Multi-modal Multi-media	50M	Mamba	AD Classification	N/A
10	DMamba [109]	arXiv24	Multi-modal Multi-media	N/A	Mamba	Reinforcement Learning	https://github.com/Toshihiro-Ota/decision-mamba
11	EMMA [110]	ICLR25	Multi-modal Multi-media	MambaV1 2.8B MambaV2 2.7B	Mamba	VQA Visual Understanding	https://github.com/xingyifei2016/EMMA
12	Ali et al. [111]	ACL25	Multi-modal Multi-media	ViM-S ViT-S	Mamba Transformer	Segmentation	https://github.com/AmeenAli/HiddenMambaAttn
13	Cobra [54]	AAAI25	Multi-modal Multi-media	Mamba 2.8B	Mamba	Visual Question Answering Closed-set Prediction	https://sites.google.com/view/cobravlm
14	Sigma [112]	WACV25	Multi-modal Multi-media	VMamba-T VMamba-S	Mamba	Multi-modal Semantic Segmentation	https://github.com/zifuwan/Sigma
15	OmniMamba [113]	arXiv25	Multi-modal Multi-media	Mamba-2 1.3B	Mamba	Understanding Generation	https://github.com/hustvl/OmniMamba
16	AVS-Mamba [114]	IEEE TMM25	Multi-modal Multi-media	N/A	Mamba	Audio-Visual Segmentation	https://github.com/SitongGong/AVS-Mamba
17	LEAF-Mamba [115]	ACM MM25	Multi-modal Multi-media	84.5M	Mamba	Salient Object Detection	N/A
18	VSumMamba [116]	ACM MM25	Multi-modal Multi-media	N/A	Mamba	Video Summarization	N/A
19	FluencyVE [117]	IEEE TMM26	Multi-modal Multi-media	4.8M	Mamba	Video Editing	https://github.com/CIMASA/FluencyVE
20	MSF-Mamba [118]	IEEE TMM26	Multi-modal Multi-media	MSF-Mamba-T 8.90M MSF-Mamba-M+ 235.32M	Mamba	Micro-Gesture Recognition	N/A

Table 6 Summary of recent SSM-based point cloud and event stream algorithms

#ID	Algorithm	Source	Domain	Parameters	Architecture	Efficiency	Code
1	PointMamba [46]	NeurIPS24	Point	Classification 12.3M Segmentation 17.4M	SSM	Classification 3.6 GFLOPs Segmentation 14.3 GFLOPs	https://github.com/LMD0311/PointMambav
2	Point Mamba [119]	arXiv24	Point	Classification 3.08M Segmentation 31.99M	SSM	Classification 1.31 GFLOPs Segmentation 55.07 GFLOPs	https://github.com/IRMVLab/Point-Mamba
3	3DMamba-Complete [120]	ACM TOMM26	Point	34.06M	Former+ Mamba	7.12 GFLOPs	N/A
4	Zubi'c et al. [44]	CVPR24	Event	S4D-ViT-B 16.5M S5-ViT-B 17.5M	Former+ SSM	N/A	N/A
5	PCM [121]	AAAI25	Point	Classification PCM-Tiny: 6.9M PCM: 34.2M Segmentation PCM-Tiny: 8.8M PCM: 40.6M	SSM	N/A	https://github.com/zhang-tao-whu/PCM
6	3DMam-balPF [122]	AAAI25	Point	N/A	Mamba	N/A	N/A
7	EventMamba (Ren) [123]	TPAMI25	Event	Classification 0.29M–3.28M Regression 0.904M	Mamba	Classification 0.219–3.722 GFLOPs Inference time 5.0–8.6 ms CPR: 5.5 ms Eye Tracking: 5.9 ms	https://github.com/rhwxmx/EventMamba
8	EventMamba (Ge) [124]	AAAI25	Event	11.21M	Mamba	Low-resolution inference 18.82 ms High-resolution inference 136.14 ms 4× faster than Transformers GPU memory reduction: 63%	https://github.com/ndwsfkba/EventMamba
9	SMamba [125]	AAAI25	Event	16.7M	Mamba	7.4 GFLOPs	N/A
10	Stru-Mamba3D [126]	ICCV25	Point	15.8M	Mamba	4.0 GFLOPs	N/A
11	PMA [127]	CVPR25	Point	1.1M–5.64M	Mamba	N/A	https://github.com/zyh16143998882/PMA
12	SI-Mamba [128]	CVPR25	Point	12.3M	Mamba	Classification 3.6 GFLOPs Segmentation 14.3 GFLOPs	https://github.com/AliBahri94/SI-Mamba
13	Mamba4D [51]	CVPR25	Point	N/A	Mamba	GPU memory reduction 87.5% Inference speed 5.36× faster	https://github.com/IRMVLab/Mamba4D
14	MambaEVT [129]	TCSVT26	Event	29.3M	Mamba	FPS: 28	https://github.com/Event-AHU/MambaEVT

Table 6 (*Continued*)

#ID	Algorithm	Source	Domain	Parameters	Architecture	Efficiency	Code
15	DAPoint-Mamba [130]	AAAI26	Point	9.571M	Mamba	5.192 GFLOPs	N/A
16	Cloud-Mamba [131]	AAAI26	Point	Classification 9.95M Part segmentation 16.57M Semantic segmentation 16.56M	Mamba	Classification 1.150 GFLOPs Part segmentation 1.234 GFLOPs Semantic segmentation 4.922 GFLOPs	https://github.com/Point-Cloud-Learning/CloudMamba

Table 7 Summary of other recent SSM-based algorithms

#ID	Algorithm	Source	Domain	Parameters	Architecture	Efficiency	Code
1	SaShiMi [36]	ICML22	Others	23.0M	S4	N/A	http://github.com/state-spaces/s4
2	Lu et al. [132]	NeurIPS23	Others	N/A	SSM	N/A	https://github.com/luchris429/s5rl
3	David et al. [133]	arXiv24	Others	N/A	SSM+HMM	N/A	N/A
4	FlashFFT-Conv [134]	ICLR24	Others	N/A	Conv+SSM	GFLOPs: 0.0565	https://github.com/HazyResearch/flash-fft-conv
5	CLDSSMs [135]	FUSION24	Others	N/A	DSSM	N/A	N/A
6	LaRocque et al. [136]	IROS24	Others	N/A	Mamba	N/A	https://github.com/norlab-ulaval/BorealTC
7	SiMBA [137]	arXiv24	Others	Small Monarch: 18.5M Base Monarch: 26.9M Large Monarch: 42M	SSM+CNN EinFFT	SiMBA-S: 3.6 GFLOPs SiMBA-B: 5.5 GFLOPs SiMBA-L: 8.7 GFLOPs	https://github.com/badripatro/Simba
8	Sharma et al. [138]	arXiv24	Others	N/A	Mamba	N/A	N/A
9	Forgione et al. [139]	CDC24	Others	N/A	SSM	N/A	N/A
10	Yang et al. [140]	arXiv24	Others	N/A	Mamba	N/A	https://github.com/nancheng58/RecMamba
11	oSpatial-Mamba [141]	SPL24	Others	1.4M	SSM	N/A	https://github.com/Audio-WestlakeU/NBSS
12	Caduceus [142]	ICML24	Others	N/A	SSM	N/A	https://github.com/kuleshov-group/caduceus
13	MambaStock [143]	arXiv24	Others	N/A	Mamba	N/A	https://github.com/zshicode/MambaStock
14	Bhirangi et al. [144]	ICML24	Others	N/A	SSM	N/A	https://hiss-csp.github.io
15	TimeMachine [145]	ECAI24	Others	N/A	Mamba	N/A	https://github.com/Atik-Ahamed/TimeMachine
16	Poli et al. [146]	ICML24	Others	70M–7B	Mechanistic Architecture Design	N/A	N/A
17	Yin et al. [147]	arXiv24	Others	N/A	SSM	N/A	N/A

Table 7 (*Continued*)

#ID	Algorithm	Source	Domain	Parameters	Architecture	Efficiency	Code
18	SST [148]	CIKM25	Others	N/A	Mamba	N/A	https://github.com/XiongxiaoXu/SST
19	PINNMamba [149]	ICML25	Others	N/A	Mamba	N/A	https://github.com/miniHuiHui/PINNMamba
20	PhyxMamba [150]	arXiv25	Others	N/A	Mamba	N/A	https://github.com/tsinghua-fib-lab/PhyxMamba
21	S/D-Mamba [151]	Neuro-computing25	Others	N/A	Mamba	N/A	https://github.com/wzhwzhwzh0921/S-D-Mamba
22	SD-SSM [152]	AAAI25	Others	N/A	SSM	N/A	https://github.com/IBM/selective-dense-state-space-model
23	Centaurus [153]	ICLR25	Others	0.378M	SSM	GFLOPs: 0.134	https://openreview.net/forum?id=PkpNRmBZ32
24	Polarization SSM [154]	ICLR25	Others	N/A	Mamba	N/A	https://github.com/VITA-Group/SSM-Bottleneck
25	PD-SSM [62]	NeurIPS26	Others	N/A	SSM	N/A	https://github.com/IBM/expressive-sparse-state-space-model
26	CompreSSM [155]	ICLR26	Others	N/A	SSM	N/A	https://github.com/camail-official/compressm

TimeMachine [145] captures global and local dependencies in multivariate forecasting. Poli et al. [146] developed a mechanistic framework for designing efficient hybrid sequence architectures. Yin et al. [147] learned digital dynamic-range compressor models from simulation prototypes.

SST [148] routes information between a Mamba global expert and a local-window Transformer. PINNMamba [149] combines subsequence-based Mamba memory with physics-informed learning, and PhyxMamba [150] uses time-delay embedding for long-horizon chaotic-system prediction. S/D-Mamba [151] models variable correlations in multivariate time series.

SD-SSM [152] uses dense selective transitions to improve expressiveness. Centaurus [153] formulates generalized SSM operations as optimizable tensor contractions for speech tasks. Polarization SSM [154] studies structural bottlenecks in associative recall. PD-SSM [62] introduces sparse structured transitions, and CompreSSM [155] progressively removes low influence states during training.

4 Experiments

In this section, an experimental comparison of five downstream tasks is presented, including single- or multi-label classification, visual object tracking, pixel-level segmentation, image-to-text generation, and person or vehicle re-identification. Representative input data are shown in Fig. 2. More details will be introduced in the following subsections.

4.1 Single- or multi-label classification

As illustrated in Fig. 3, we conduct experiments on three categories of vision tasks. Figure 3(a), (b) and (c) present the performance of video-based recognition on the Kinetics-400, HMDB-51/UCF-101 and LVU datasets, respectively. For the single-label classification problem, we calculate the accuracy of recent works on the widely used ImageNet-1K [2] dataset. As shown in Fig. 3(d), we found that the base version of VMamba [43] and Mamba-2D [156] achieve better results on the ImageNet1K dataset, i.e., 83.2% and 83.0% on the top-1 accuracy, respectively. We also note that current Mamba-based vision models are all tiny, small, or base versions, and seldom pre-train a large or huge version of the Mamba network. The overall performance is comparable to some Transformer-based models, but still inferior to the state-of-the-art models on the ImageNet classification dataset.

For the multi-label classification, we select the pedestrian attribute recognition (PAR) task [6][3] and conduct experiments on the PA100K [157] and PETA [158] datasets. The PA100K dataset contains 100,000 samples collected from 598 scenarios and involves 26 pedestrian attributes. We split the training, validation, and testing subsets based on default settings (8:1:1). The PETA dataset involves 61 binary attributes and 19,000 person images. The training, validation, and testing subsets contain 9500, 1900, and

[3] https://github.com/wangxiao5791509/Pedestrian-Attribute-Recognition-Paper-List.

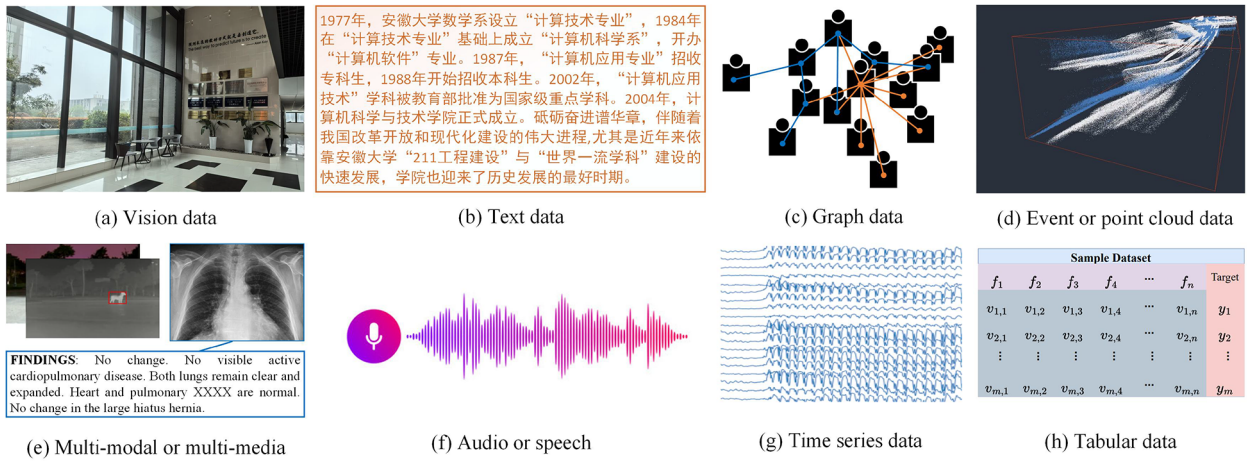


Figure 2 Representative input data that can be processed using the SSM

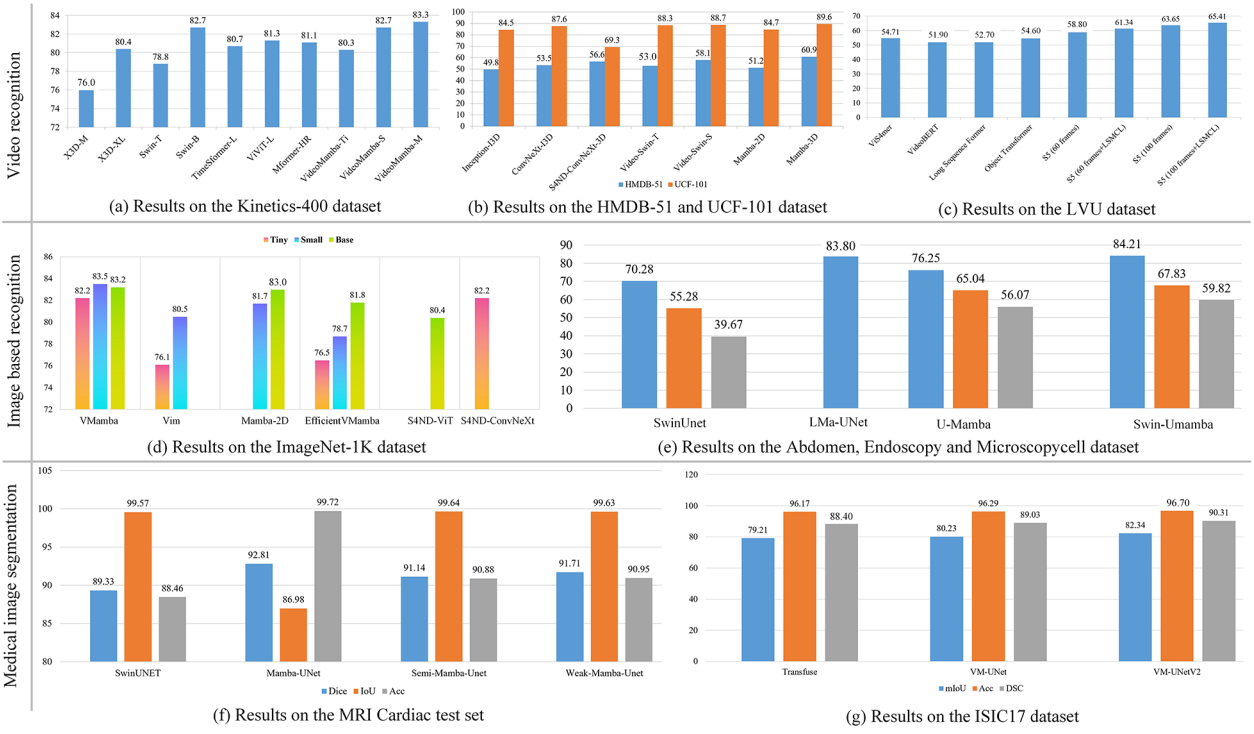


Figure 3 Experimental results on video-based recognition (a, b, c), image-based recognition (d), medical image-based segmentation (e, f, g)

7600 images, respectively. By following its default settings, 35 pedestrian attributes were selected for the experiment.

The ViT-S [18] and Mamba-based network Vim-S [45] were adopted as the backbone for this experiment. We follow the vision-language fusion-based PAR framework VTB [159], which takes the pedestrian image and attribute set as input and predicts the logistic scores of each attribute. From the experimental results reported in Table 8, we found that the Vim-S based PAR model achieved 81.08/73.75/80.91/84.96/82.52 on the PETA dataset, and 80.41/78.03/85.39/88.37/86.39 on the PA100K dataset. While these results demonstrate the strong performance of SSMs, they are still significantly inferior to those of the PAR algorithms developed based on the Transformer network. For example, the ViT-B based VTB achieved

Table 8 Comparison with state-of-the-art methods on the PETA and PA100K datasets. The best and second-best results are shown in bold and italic, respectively. "-" means this indicator is not available. VTB* indicates that VTB uses CLIP's feature extractor

Method	Source	Backbone	PETA					PA100K				
			mA	Acc	Prec	Recall	F1	mA	Acc	Prec	Recall	F1
JLAC [160]	AAAI 2020	ResNet50	86.96	80.38	87.81	87.09	87.50	82.31	79.47	87.45	87.77	87.61
SCRL [161]	TCSVT 2020	ResNet50	87.20	N/A	*89.20*	87.50	88.30	80.60	N/A	88.70	84.90	82.10
SSCsoft [162]	ICCV 2021	ResNet50	86.52	78.95	86.02	87.12	86.99	81.87	78.89	85.98	89.10	86.87
IAA-Caps [163]	PR 2022	OSNet	85.27	78.04	86.08	85.80	85.64	81.94	80.31	88.36	88.01	87.80
MCFL [164]	NCA 2022	ResNet-50	86.83	78.89	84.57	88.84	86.65	81.53	77.80	85.11	88.20	86.62
DRFormer [165]	NC 2022	ViT-B/16	**89.96**	81.30	85.68	**91.08**	88.30	82.47	80.27	87.60	88.49	88.04
VAC-Combine [166]	IJCV 2022	ResNet50	N/A	N/A	N/A	N/A	N/A	82.19	80.66	88.72	88.10	88.41
DAFL [167]	AAAI 2022	ResNet50	87.07	78.88	85.78	87.03	86.40	83.54	80.13	87.01	89.19	88.09
CGCN [168]	TMM 2022	ResNet	87.08	79.30	83.97	89.38	86.59	N/A	N/A	N/A	N/A	N/A
CAS-SAL-FR [169]	IJCV 2022	ResNet50	86.40	79.93	87.03	87.33	87.18	82.86	79.64	86.81	87.79	85.18
PromptPAR [170]	TCSVT 2025	ViT-L/14	*88.76*	*82.84*	89.04	89.74	*89.18*	**87.47**	*83.78*	*89.27*	**91.70**	**90.15**
SequencePAR [171]	PR 2026	ViT-L/14	N/A	**84.92**	**90.44**	*90.73*	**90.46**	N/A	**83.94**	**90.38**	90.23	*90.10*
VTB [159]	TCSVT 2022	ViT-B/16	85.31	79.60	86.76	87.17	86.71	83.72	80.89	87.88	89.30	88.21
VTB* [159]	TCSVT 2022	ViT-L/14	86.34	79.59	86.66	87.82	86.97	*85.30*	81.76	87.87	*90.67*	88.86
VTB [159]	TCSVT 2022	ViT-S	82.51	77.23	85.75	84.95	85.01	78.76	77.61	87.41	85.35	85.94
Vim-PAR		Vim-S	81.08	73.75	80.91	84.96	82.52	80.41	78.03	85.39	88.37	86.39

85.31/79.60/86.76/87.17/86.71 on the PETA dataset, and 83.72/80.89/87.88/89.30/88.21 on the PA100K dataset.

4.2 Visual object tracking

In this subsection, we compare the Mamba with the Transformer and CNN-based backbone for the tracking task[4] based on OSTrack [172]. Specifically, the CNN based trackers are TrDiMP [173], ToMP50 [174], DiMP50 [175], PrDiMP [176], KYS [177], and ATOM [178]; the Transformer based trackers are HDETrack [179], AiATrack [180], STARK [181], TransT [182], MixFormer [183], and SimTrack [184]. To achieve a fair comparison, we train and test these trackers on a large-scale event-based tracking dataset, EventVOT [179], which contains 841, 18, and 282 videos, respectively. The detailed experimental results are reported in Table 9. Note that three widely used evaluation metrics are used for the comparison, including success rate (SR), precision rate (PR), and normalized precision rate (NPR). According to Table 9, we found that replacing the ViT with the Mamba backbone network slightly decreases the performance, but it greatly reduces the number of parameters (4.1M only). Therefore, we draw the conclusion that the Mamba network will be a promising choice for event-based tracking.

Table 9 Comparison between different trackers on the EventVOT dataset. SR: success rate; PR: precision rate; NPR: normalized precision rate

Trackers	Source	Backbone	SR	PR	NPR	Params (M)	FPS
TrDiMP [173]	CVPR21	ResNet50	39.9	34.8	48.7	26.3	26
ToMP50 [174]	CVPR22		37.6	32.8	47.4	26.1	25
DiMP50 [175]	ICCV19		52.6	51.1	67.2	26.1	43
PrDiMP [176]	CVPR20		55.5	57.2	70.4	26.1	30
KYS [177]	ECCV20		38.7	37.3	49.8	N/A	20
ATOM [178]	CVPR19		44.4	44.0	57.5	8.4	30
HDETrack [179]	CVPR24	ViT	57.8	62.2	73.5	92.1	105
AiATrack [180]	ECCV22		57.4	59.7	72.8	15.8	38
STARK [181]	ICCV21		44.5	39.6	55.7	28.1	42
TransT [182]	CVPR21		54.3	56.5	68.8	18.5	50
MixFormer [183]	CVPR22		49.9	49.6	63.0	35.6	25
SimTrack [184]	ECCV22		55.4	57.5	69.9	57.8	40
OSTrack [172]	ECCV22	ViT-B	55.4	60.4	71.1	92.1	105
		ViT-S	52.0	53.2	66.8	54.3	109
		Vim-S	55.6	59.1	70.4	4.1	41

4.3 Pixel-level segmentation

Recently, the Mamba network has been increasingly explored in medical image segmentation, as illustrated in Fig. 3 (e)-(g), owing to its efficient long-range dependency modeling and favorable computational complexity. For example, the Swin Transformer-based model SwinUNet [185] achieves 89.33/99.57/88.46 in terms of Dice, IoU, and accuracy, respectively, on the MRI Cardiac dataset, demonstrating the effectiveness of Transformer-based architectures for capturing global contextual information. In comparison, Mamba-based U-shaped architectures achieve comparable or even superior segmentation performance while benefiting from the efficient sequence modeling capability of the SSMs. Representative methods include Mamba-UNet [186], Semi-Mamba-UNet [187], and Weak-Mamba-UNet [61], which extend the Mamba architecture to fully supervised, semi-supervised, and weakly supervised segmentation settings, respectively.

[4] https://github.com/wangxiao5791509/Single_Object_Tracking_Paper_List.

Table 10 Comparison between the performance of R2Gen-GPT-Vim-Small and other methods on the IU-Xray dataset. R2Gen-GPT-Vim-S* and R2GenGPT-Vim-S denote Vim-S is initialized with and without pre-trained parameters, respectively

Method	Backbone	CIDEr	BLEU-4	ROUGE-L
R2Gen [190]	CNN	0.398	0.165	0.371
KERP [191]	CNN	0.280	0.162	0.339
HRGP [192]	CNN	0.343	0.151	0.322
MKG [193]	CNN	0.304	0.147	0.367
PPKED [194]	CNN	0.351	0.168	0.376
MGSK [195]	CNN	0.382	0.178	0.381
CA [196]	ResNet-50	N/A	0.169	0.381
CMCL [197]	CNN	N/A	0.162	0.378
DCL [198]	CNN	0.586	0.163	0.383
R2GenGPT	Swin-B	0.524	0.152	0.352
R2GenGPT	Vim-S	0.388	0.152	0.355
R2GenGPT	Vim-S*	0.382	0.171	0.371

4.4 Image-to-text generation

For the image-to-text generation, we select the X-ray report generation task, which takes the X-ray medical image as the input and generates the medical report.[5] For the experiment, we select the R2GenGPT[6] as our baseline and evaluate its performance on the IU-Xray dataset [188]. R2GenGPT consists of a visual encoder (Swin Transformer [19]), a linear layer, and a large language model (llama-2-7B-chat [189]). The training approach involves freezing the language model initially and subsequently fine-tuning the visual encoder and the linear layer. We replaced the Swin Transformer with the Vim model [45] and compared the results with other methods in Table 10. As both models utilize pre-trained components, Vision Mamba demonstrates superior performance over the Swin Transformer model in terms of BLEU-4 and ROUGE-L scores.

4.5 Person or vehicle re-identification

As shown in Table 11, we conduct experiments on two re-identification (re-ID) tasks, i.e., the person re-identification [221] and vehicle re-identification [220]. For the person re-ID task, four widely used datasets are used, including MSMT17 [222], Market1501 [223], DukeMTMC [224], and Occluded-Duke [213]. These datasets are captured from different scenes, and the samples are collected from surveillance systems with overlapping coverage of cameras, which have challenges such as cross-time span, occlusion, and background interference. For the vehicle re-ID, VeRi-776 [225] and VehicleID [226] datasets are utilized for the experimental validation. Different from pedestrian samples, the change of observation viewpoints also brings significant appearance differences for vehicles. Thus, the vehicle datasets are additionally provided with viewpoint labels to mark the different viewpoints of the vehicle samples. For the above datasets, we use the cumulative matching characteristic (CMC) curve and mean average precision (mAP) as evaluation metrics.

Referring to mainstream frameworks such as TransReID [219] and Strong Baseline [227], we retained ID Loss, Triplet Loss, and BN Layer, and replaced the CNN and Transformer backbones using Vim [45] and VMamba [43] to explore the potential of Mamba for re-identification tasks, and the compared results are shown in Table 11. The selective scanning mechanism (i.e., the SSM) proposed by the Mamba model allows for sequence modeling with low complexity, and Vim and VMamba build on it by proposing an SSM modeling approach for 2D image data. Compared to CNN-based models that require complex module design, the simple Mamba network is already effective. Even compared with models with high complexity, such as ViT [18], the bidirectional scanning mechanism proposed by Vim has fewer training parameters, and it shows effectiveness on the VehicleID dataset. In contrast, VMamba's cross-scanning mechanism, which does not rely on the Transformer's structure (e.g., position embedding and class token), has achieved comparable results on these datasets. For this reason, we expect more Mamba-based studies applicable to the re-identification task in the future.

5 Discussion

5.1 Systematic comparison with transformers

Although SSMs are increasingly viewed as potential alternatives to Transformers, the two architecture families differ fundamentally in how they represent, propagate, and retrieve information. Transformers explicitly model interactions between token pairs, whereas SSMs compress historical information into a latent state and propagate it recurrently. These distinct mechanisms lead to different trade-offs in modeling principles, long-range dependency capture, computational complexity, representational capacity, and application suitability.

Modeling mechanisms Transformers rely on self-attention, which explicitly computes content dependent interactions between tokens:

$$\text{Attention}(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) = \text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^{\top}}{\sqrt{d}}\right)\boldsymbol{V}, \tag{8}$$

where $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ denote the query, key, and value matrices, respectively, and d is the feature dimension. This operation allows each token to directly retrieve information from other positions according to pairwise similarity. In contrast, an SSM summarizes previous inputs through

[5] https://github.com/Event-AHU/Medical_Image_Analysis.

[6] https://github.com/wang-zhanyu/R2GenGPT.

Table 11 Comparison with methods based on CNN and Transformer on person re-identification and vehicle re-identification datasets

Backbone	Method	MSMT17		Market1501		DukeMTMC		Occluded-Duke		Method	VeRi-776		VehicleID	
		mAP	R1	mAP	R1	mAP	R1	mAP	R1		mAP	R1	R1	R5
CNN	CBN [199]	42.9	72.8	77.3	91.3	67.3	82.5	N/A	N/A	PRReID [200]	72.5	93.3	72.6	88.6
	OSNet [201]	52.9	78.7	84.9	94.8	73.5	88.6	N/A	N/A	SAN [202]	72.5	93.3	79.7	94.3
	MGN [203]	52.1	76.9	86.9	95.7	78.4	88.7	N/A	N/A	UMTS [204]	75.9	95.8	80.9	87.0
	RGA-SC [205]	57.5	80.3	88.4	96.1	N/A	N/A	N/A	N/A	VANet [206]	66.3	89.8	83.3	96.0
	SAN [207]	55.7	79.2	88.0	96.1	75.7	87.9	N/A	N/A	SPAN [208]	68.9	94.0	N/A	N/A
	SCSN [209]	58.5	83.8	88.5	95.7	79.0	91.0	N/A	N/A	PGAN [210]	79.3	96.5	78.0	93.2
	ABDNet [211]	60.8	82.3	88.3	95.6	78.6	89.0	N/A	N/A	PVEN [212]	79.5	95.6	84.7	97.0
	PGFA [213]	N/A	N/A	76.8	91.2	65.5	82.6	37.3	51.4	SAVER [214]	79.6	96.4	79.9	95.2
	HOReID [215]	N/A	N/A	84.9	94.2	75.6	86.9	43.8	55.1	CFVMNet [216]	77.1	95.3	81.4	94.1
	ISP [217]	N/A	N/A	88.6	95.3	80.0	89.6	52.3	62.8	GLAMOR [218]	80.3	96.5	78.6	93.6
Transformer	DeiT-B/16 [219]	61.4	81.9	86.6	94.4	78.9	89.3	53.1	60.6	DeiT-B/16 [219]	78.4	95.9	83.1	96.8
	ViT-B/16 [219]	61.0	81.8	86.8	94.7	79.3	88.8	53.1	60.5	ViT-B/16 [219]	78.2	96.5	82.3	96.1
	VehicleMAE [220]	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	VehicleMAE [220]	85.6	97.9	N/A	N/A
Mamba	Vim-T/16	40.1	62.6	75.7	89.4	66.5	81.8	35.4	45.1	Vim-T/16	62.9	89.2	67.0	88.2
	Vim-S/16	42.2	66.2	77.5	89.7	67.4	83.0	40.8	51.3	Vim-S/16	61.6	89.6	78.2	94.8
	VMamba-T/16	51.0	75.6	83.3	92.8	74.9	87.3	49.4	58.3	VMamba-T/16	77.3	95.9	78.5	93.5
	VMamba-B/16	51.1	75.3	84.3	93.2	77.4	88.0	48.1	57.4	VMamba-B/16	77.5	95.6	82.5	96.1

a latent state:

$$\begin{aligned} \boldsymbol{h}_t &= \overline{\boldsymbol{A}}_t \boldsymbol{h}_{t-1} + \overline{\boldsymbol{B}}_t \boldsymbol{x}_t, \\ \boldsymbol{y}_t &= \boldsymbol{C}_t \boldsymbol{h}_t. \end{aligned} \tag{9}$$

Classical structured SSMs mainly use input independent transition dynamics, while selective SSMs introduce input dependent parameters to improve content adaptivity. Nevertheless, they still propagate information through a compressed state rather than constructing a complete token-to-token interaction matrix. Thus, Transformers prefer explicit information routing, whereas SSMs emphasize efficient state-based propagation.

Long-range dependency modeling Self-attention provides a direct interaction path between any two tokens, regardless of their distance. This property makes Transformers effective for global comparison, and context-dependent reasoning, but the cost of maintaining all pairwise interactions increases rapidly with sequence length. SSMs instead preserve long-range information by repeatedly updating a latent state. Structured transitions, stable initialization, and convolutional formulations allow SSMs to retain information over long horizons, while selective mechanisms determine which information should be preserved or discarded. This enables efficient long-sequence processing, but the finite-dimensional state may become an information bottleneck when exact historical tokens must be recovered. Consequently, Transformers are generally stronger in precise retrieval and pairwise comparison, whereas SSMs are well suited to long-term signal propagation and continuous sequential data.

Computational and memory complexity For a sequence of length L, standard self-attention requires $\mathcal{O}(L^2)$ computation and memory for pairwise attention scores. By comparison, recurrent SSM processing has linear complexity in L when the state and feature dimensions are fixed. During inference, each layer only updates a fixed-size state instead of storing all previous key and value representations. Hardware-aware scan algorithms further enable parallel training while maintaining linear sequence complexity. However, asymptotic complexity does not fully determine practical speed. SSM performance depends on the state dimension, scan implementation, memory access pattern, and hardware support. For short or moderate sequences, highly optimized Transformer implementations may remain equally efficient or faster.

Representation capability Self-attention can dynamically construct explicit relationships between arbitrary tokens. This is particularly useful for exact retrieval, pairwise matching, in-context learning and multi-modal alignment. Multi-head attention can also represent multiple relationships simultaneously. SSMs encode historical context in a continuously updated state, which naturally supports temporal continuity, causal processing, and streaming inference. Structured SSMs provide useful inductive biases for dynamical systems and regularly sampled signals, while selective SSMs improve input-dependent memory. However, their finite-dimensional state may lose detailed historical information and does not explicitly reproduce the complete pairwise interaction structure of self-attention.

Application scenarios Transformers are generally preferable when explicit global interactions, precise information

retrieval, contextual reasoning, and cross-modal alignment are essential. Through self-attention, they can directly model relationships between arbitrary tokens, making them particularly effective for question answering, in-context learning, information retrieval, and multi-modal understanding. By contrast, SSMs are particularly well suited to very long sequences, continuous or streaming data, autoregressive inference, high-resolution inputs, and memory constrained deployment. Their linear sequence complexity and recurrent state representation enable efficient long-range processing and incremental inference. Representative applications include audio and speech processing, time-series forecasting, genomic sequence modeling, event-stream analysis, long-video understanding, and online prediction. Therefore, Transformers are often preferred for intensive tasks, whereas SSMs offer clearer advantages in long-contexts, streaming, and resource-constrained scenarios.

5.2 Rationale for downstream task selection

The selected downstream tasks are designed to evaluate the performance of SSM architectures across a variety of challenging scenarios. Image classification examines global representation learning and long-range contextual aggregation. Visual object tracking evaluates temporal state propagation and sequential feature updating. Pixel-level segmentation assesses the ability to combine global context with fine-grained spatial information. Image-to-text generation evaluates cross-modal alignment and autoregressive sequence modeling. Person or vehicle re-identification examine fine-grained discriminative learning and global feature matching across different views. Collectively, these tasks cover image-level prediction, temporal modeling, dense prediction, multi-modal generation, and retrieval-based recognition. They also include both discriminative and generative settings, thereby providing a broad and complementary evaluation of the representational and computational characteristics of the SSM-based models.

5.3 Limitations of current SSM-based models

Despite their efficiency in long-sequence modeling, current SSM-based models still exhibit several limitations. First, compressing history into a finite-dimensional state may create an information bottleneck when exact past details are required. Second, the lack of explicit pairwise token interactions can limit precise retrieval and relational reasoning. Third, practical efficiency depends on optimized kernels and hardware support, so linear complexity does not always ensure faster execution. Finally, many recent SSM variants still require broader and more standardized evaluation across tasks, scales, and hardware platforms.

6 Challenge and opportunity

The SSM has been widely studied and applied in many applications. However, it should be noted that research in this area is still in the early stages. This paper presents several research points that deserve attention.

Current SSMs model still inferior to the mainstream transformer networks From the experimental results reported in Sect. 4, we found that there is still room for performance improvement in SSM-based models. The SSMs pre-trained on large-scale datasets, such as ImageNet [2], play a critical role in many downstream tasks. However, the base, large, and huge versions of SSMs are seldom released. We believe this may be an obstacle to high performance on the CV tasks.

Further exploration and research are needed to understand the advantages of SSMs for GPU usage According to our experiments, the memory consumption is lower or comparable to the Transformer networks on some downstream tasks. A significant improvement in this aspect can be observed, but some tasks are not. The study on mining the lower GPU memory consumption is worth further exploration and research.

Its advantages in long-term vision data or high-resolution Since the SSM architecture significantly reduces the complexity of the model theoretically, its modeling capability on high-resolution data (remote sensing data or X-ray medical images) or long-term sequence data (long-term video frames) is of great value. However, these aspects are not addressed well using other strong models like the Transformer network.

Pre-trained large models using the SSM architecture In the pre-trained large model era, the scaling of deep neural networks is an important step for general artificial intelligence. Current large models are built based on CNN or Transformer networks, and a few of them adopt the SSM architecture. Recently, Jamba [228], released by AI21Labs, is a novel large language model built by fusing the Transformer, Mamba, and mixture of experts (MoE). It supports the input of context length up to 256K tokens and also achieves comparable performance with Mixtral-8x7B [229] and Llama-2 70B [189]. The study on building pure Mamba or hybrid architectures will be a promising direction for pre-trained large models.

Multi-modal learning using the SSM architecture Early multi-modal related works focused on how to learn modality-specific and modality-shared representations. Influenced by the Transformer network, current multi-modal algorithms usually directly encode and fuse the multiple cues in a unified Transformer network [230, 231]. Thus,

the cost of the inference phase may be twice compared of a single modality only. How to design new SSM-based backbones for cost-sensitive multi-modal learning is an important research topic.

Developing novel scan operators for the SSMs The scan is a key operator for the SSM architecture, and the 1D and 2D data are usually processed with different scan mechanisms. For example, VMamba [43] scans an image using CSM (scan expand) and merges the four output features as the final 2D feature map. To deal with more specialized remote sensing data, some researchers have proposed additional scanning mechanisms to capture skewed feature representations to obtain more comprehensive features [232]. Therefore, it is natural to design novel scan schemes to enhance the feature learning of SSMs. For example, it is possible to develop new track-changing scan methods to better encode the point cloud and event streams.

The potential for generalization in SSMs Compared with CNNs and Transformers, which suffer from limited receptive fields or high computational complexity, SSMs offer linear complexity and global receptive fields, showing strong potential for domain generalization. However, current SSM-based networks illustrate limited domain generalization ability, as noted in DGMamba [233]. Long et al. [233] attempted to address this issue from the perspective of hidden states and inappropriate scan mechanisms by proposing the hidden state suppressing (HSS) and semantic-aware patch refining (SPR) strategies. We believe more insights and improvements can be conducted to further improve the overall performance of domain generalization.

The applicability of the Mamba mechanism to vision tasks A recent study by Yu and Wang [234] provided a conceptual analysis of the Mamba mechanism, demonstrating its strong suitability for tasks involving long sequences and autoregressive properties. This presents a promising opportunity to explore its applications in domains with these characteristics, such as object detection and semantic segmentation. However, it also poses a significant challenge: adapting or developing alternative algorithms for tasks lacking these traits, such as image classification, where the mechanism may be less inherently compatible.

Using the latest SSM model to empower the recent deep neural network model In the early stage of the third wave of deep learning, many innovative neural network modules or designs are proposed. For example, knowledge distillation, pyramid structure, network in network [235], diffusion models, and the generative adversarial network (GAN). Enhancing the SSM based on these successful modules or introducing SSM into these modules has the potential to yield enhanced performance.

7 Conclusion

In this paper, we provide a comprehensive review of extant research on the SSM to facilitate the rapid comprehension of the most recent advancements in the field. First, we provide an introduction to the origin of the SSM. Then, we delve into the main streams of SSMs and review these works from the perspectives of the origin algorithm, natural language processing, computer vision, graphs, multimodal and multi-media, event and point data , and other domains. Due to the applications of SSMs in each domain still being in the early stages, in this paper, we conduct experiments on multiple research problems, such as single- or multi-label classification, visual object tracking, pixel-level segmentation, image-to-text generation and person or vehicle Re-ID. From the experimental results, we found that current SSMs achieve similar performance to some Transformer networks. However, the overall results are still inferior to the state-of-the-art models. Moreover, the decrease in memory usage can be observed in the downstream tasks. Then, we summarize the challenges still existing in the SSMs and propose several research opportunities to help researchers better understand this direction.

Acknowledgements
The authors acknowledge the High-performance Computing Platform of Anhui University for providing computing resources.

Author contributions
All authors contributed to the conception and design of the study. XW, SW, YD, YL, WW, YR, WK, JH, SL, HY, and ZW drafted the manuscript and contributed to the collection and analysis of the literature. BJ, CL, YW, YT, and JT critically reviewed and revised the manuscript. All authors read and approved the final manuscript.

Funding information
This work was supported by the National Natural Science Foundation of China (No. 62102205) and the Anhui Provincial Natural Science Foundation-Outstanding Youth Project (No. 2408085Y032). The funding bodies had no role in the design of the study; in the collection, analysis, or interpretation of data; or in writing the manuscript.

Data availability
The authors confirm that no new datasets were generated during this study. All datasets used and analyzed in the experiments are publicly available from their original sources, as cited in the manuscript.

Declarations

Competing interests
The authors declare that they have no competing interests.

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Received: 15 June 2026 Revised: 22 September 2026
Accepted: 23 September 2026 Published online: 21 October 2026

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